Sections: B,C & D

Linear Algebra (MT-1004)

Homework#35

Instructor: Muhammad Ali

Question 1

Consider the following matrices

$$A = \begin{bmatrix} 6 & 2 & 1 \\ 2 & 6 & 2 \\ 1 & 2 & 6 \end{bmatrix}.$$

$$B = \begin{bmatrix} 4 & 1 & 2 \\ 1 & 4 & 1 \\ 2 & 1 & 4 \end{bmatrix}$$

- Find the eigenvalues of the matrices A and B
- Find the eigenvectors corresponding to each eigenvalue.
- Check if the matrices are diagonalizable.
- Perform the orthogonal diagonalization of A and B.
- Write A and B as sum of Rank 1 projection matrices.